



ASSESSMENT COVERSHEET

Attach this coversheet as the cover of your submission. All sections must be completed.

Section A: Submission Details

Programme : Bachelor Of Information Technology (Hons) In Computer System Security
Course Code & Name : IKB21403 - VULNERABILITY ANALYSIS
Course Lecturer(s) : Nor Adani Kamal Mohamad Nasir
Submission Title : FILE ANALYSIS – SCANNING RESULT
Deadline : Day 30 Month 04 Year 2026 Time 11.59pm
Penalties :

- 5% will be deducted per day to a maximum of four (4) working days, after which the submission will **not** be accepted.
- Plagiarised work is an Academic Offence in University Rules & Regulations and will be penalised accordingly.

Section B: Academic Integrity

Tick (✓) each box below if you agree:

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|-------------------------------------|--|
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| ☒ | This submission is my own, unless indicated with proper referencing. |
| ☒ | This submission has not been previously submitted or published. |
| ☒ | This submission follows the requirements stated in the course. |

Section C: Submission Receipt

(must be filled in manually)

Office Receipt of Submission

Date & Time of Submission (stamp)	Student Name(s)	Student ID(s)
	1 HAZIQ DANIAL BIN NOR AZAN	52215225213

Student Receipt of Submission

This is your submission receipt, the only accepted evidence that you have submitted your work. After this is stamped by the appointed staff & filled in, cut along the dotted lines above & retain this for your record.

Date & Time of Submission (stamp)	Course Code	Submission Title	Student ID(s) & Signature(s)

Question 1: Analyse packet1.pcap and find the flag.

1. Open the .pcap file

- Launch Wireshark
- Open packet1.pcap

2. Look for suspicious traffic

- Most packets are ICMP (ping)

3. Follow the ICMP stream

In Wireshark:

1. Click any ICMP packet
2. Right-click → Follow → ICMP Stream

The image shows a Wireshark packet capture analysis of packet1.pcap. The top pane displays a list of 41 ICMP Echo (ping) packets. The middle pane shows the details of the selected packet (Frame 37), including Ethernet II, Internet Protocol Version 4, and Internet Control Message Protocol (ICMP) fields. The bottom pane shows the raw data (hex and ASCII) of the selected packet, which is a Base64 encoded string: U1VDVEYyMDIze2 FpX2IzX2 Nvb2x9.

No.	Time	Source	Destination	Protocol	Length	Info
22	10.012579	171.64.20.62	172.24.20.31	ICMP	98	Echo (ping) reply id=8x7a93, seq=11/2816, ttl=64 (request in 21)
23	11.014447	172.24.20.31	171.64.20.62	ICMP	98	Echo (ping) request id=8x7a93, seq=12/3072, ttl=63 (reply in 24)
24	11.014466	171.64.20.62	172.24.20.31	ICMP	98	Echo (ping) reply id=8x7a93, seq=12/3072, ttl=64 (request in 23)
25	12.016197	172.24.20.31	171.64.20.62	ICMP	98	Echo (ping) request id=8x7a93, seq=13/3328, ttl=63 (reply in 26)
26	12.016124	171.64.20.62	172.24.20.31	ICMP	98	Echo (ping) reply id=8x7a93, seq=13/3328, ttl=64 (request in 25)
27	13.017863	172.24.20.31	171.64.20.62	ICMP	98	Echo (ping) request id=8x7a93, seq=14/3584, ttl=63 (reply in 28)
28	13.017885	171.64.20.62	172.24.20.31	ICMP	98	Echo (ping) reply id=8x7a93, seq=14/3584, ttl=64 (request in 27)
29	14.019271	172.24.20.31	171.64.20.62	ICMP	98	Echo (ping) request id=8x7a93, seq=15/3840, ttl=63 (reply in 30)
30	14.019288	171.64.20.62	172.24.20.31	ICMP	98	Echo (ping) reply id=8x7a93, seq=15/3840, ttl=64 (request in 29)
31	15.020536	172.24.20.31	171.64.20.62	ICMP	98	Echo (ping) request id=8x7a93, seq=16/4096, ttl=63 (reply in 32)
32	15.020554	171.64.20.62	172.24.20.31	ICMP	98	Echo (ping) reply id=8x7a93, seq=16/4096, ttl=64 (request in 31)
33	16.022411	172.24.20.31	171.64.20.62	ICMP	98	Echo (ping) request id=8x7a93, seq=17/4352, ttl=63 (reply in 34)
34	16.022431	171.64.20.62	172.24.20.31	ICMP	98	Echo (ping) reply id=8x7a93, seq=17/4352, ttl=64 (request in 33)
35	17.023677	172.24.20.31	171.64.20.62	ICMP	98	Echo (ping) request id=8x7a93, seq=18/4608, ttl=63 (reply in 36)
36	17.023696	171.64.20.62	172.24.20.31	ICMP	98	Echo (ping) reply id=8x7a93, seq=18/4608, ttl=64 (request in 35)
37	18.025629	172.24.20.31	171.64.20.62	ICMP	98	Echo (ping) request id=8x7a93, seq=19/4864, ttl=63 (reply in 39)
38	18.025681	172.24.20.31	171.64.20.62	ICMP	98	Echo (ping) request id=8x7a93, seq=19/4864, ttl=63 (reply in 39)
39	18.025699	171.64.20.62	172.24.20.31	ICMP	98	Echo (ping) reply id=8x7a93, seq=19/4864, ttl=64 (request in 38)
40	19.026884	172.24.20.31	171.64.20.62	ICMP	98	Echo (ping) request id=8x7a93, seq=20/5120, ttl=63 (reply in 41)
41	19.026985	171.64.20.62	172.24.20.31	ICMP	98	Echo (ping) reply id=8x7a93, seq=20/5120, ttl=64 (request in 40)

Frame 37: Packet, 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface 0
Ethernet II, Src: PaloAltoNetw_ea:40:10 (b4:0c:25:ea:40:10), Dst: VMware_58:4f:6e (00:0c:29:58:4f:6e)
Internet Protocol Version 4, Src: 172.24.20.31, Dst: 171.64.20.62
Internet Control Message Protocol
Type: Echo (ping) reply (0)
Code: 0
Checksum: 0x1ca8 [correct]
[Checksum Status: Good]
Identifier (BE): 1 (0x0001)
Identifier (LE): 256 (0x0100)
Sequence Number (BE): 0 (0x0000)
Sequence Number (LE): 0 (0x0000)
Data (28 bytes)
0000 00 0c 29 58 4f 6e b4 0c 25 ea 40 10 08 00 45 00 ...XOn..%@...E
0010 00 38 02 2d 00 00 3f 01 f9 e2 ac 18 14 1f ab 40 ...8...?.....@
0020 14 2e 00 00 1c a8 00 01 00 00 55 31 50 44 50 45 ...>.....UVDVE
0030 53 79 40 44 49 7a 68 32 48 70 58 32 6c 7a 58 32 ...YMDIze2 FpX2IzX2
0040 4e 76 62 32 78 39 ...Nvb2x9

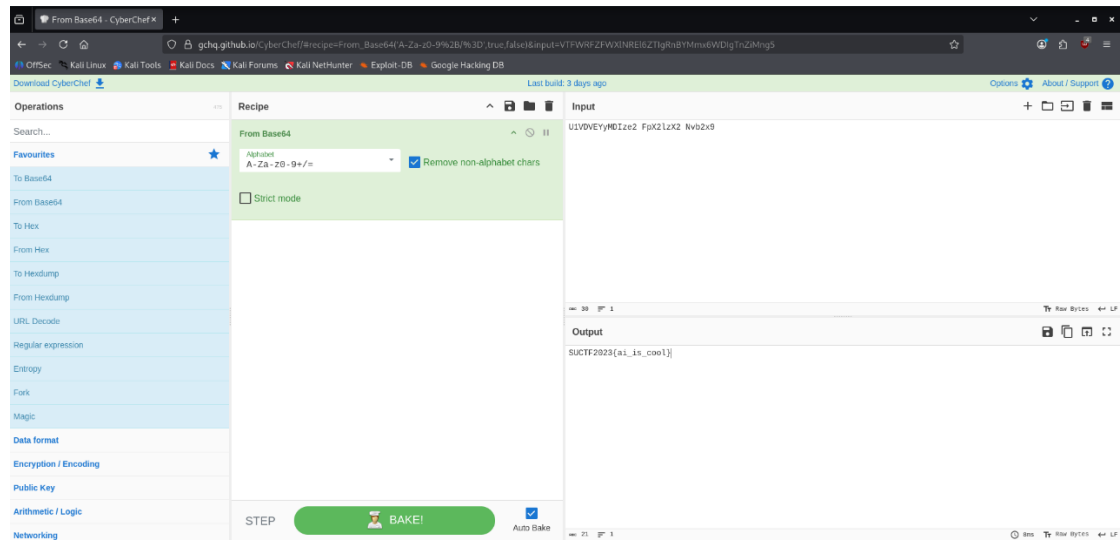
4. Check packet payload

In the bottom pane:

- Look at **Data / Payload** section
- You'll see something like:
U1VDVEYyMDIze2 FpX2IzX2 Nvb2x9
- This is **Base64 encoded text**

5. Go to website CyberChef and use FROM BASE64

- Paste the encoded text **U1VDVEYyMDIze2 FpX2lzX2 Nvb2x9**



- We get the output is **SUCTF2023{ai_is_cool}**
- Decode: **SUCTF2023{ai_is_cool}**
- Flag: **SUCTF2023{ai_is_cool}**

Question 2: Analyse packet2.pcap and find the flag.

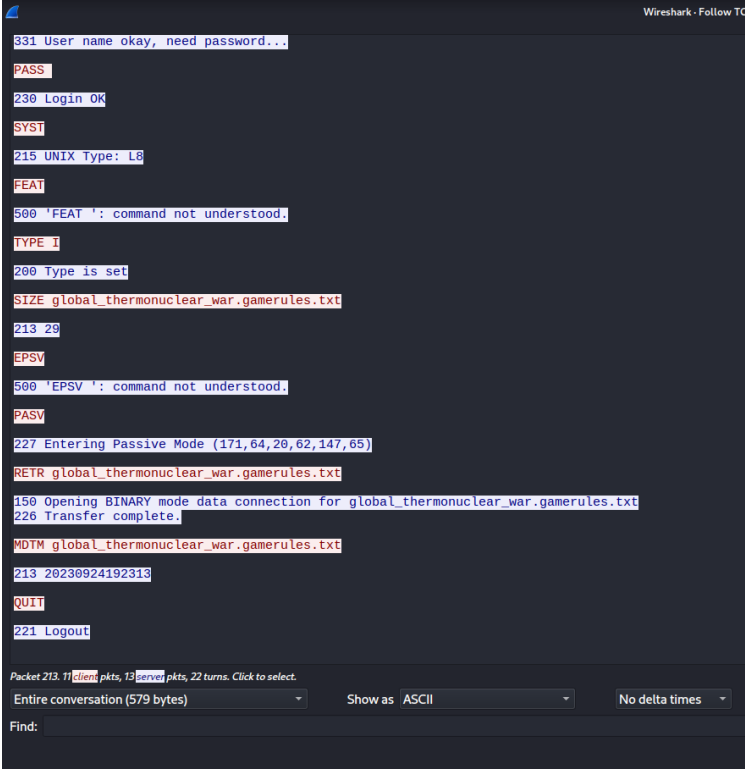
1. Identify the protocol

- You used Follow TCP Stream
- You see commands like:

USER

PASS

RETR global_thermonuclear_war.gamerules.txt



```
331 User name okay, need password...
PASS
230 Login OK
SYST
215 UNIX Type: L8
FEAT
500 'FEAT ': command not understood.
TYPE I
200 Type is set
SIZE global_thermonuclear_war.gamerules.txt
213 29
EPSV
500 'EPSV ': command not understood.
PASV
227 Entering Passive Mode (171,64,20,62,147,65)
RETR global_thermonuclear_war.gamerules.txt
150 Opening BINARY mode data connection for global_thermonuclear_war.gamerules.txt
226 Transfer complete.
MDTM global_thermonuclear_war.gamerules.txt
213 20230924192313
QUIT
221 Logout
```

Packet 213: 11 client pkts, 13 server pkts, 22 turns. Click to select.

Entire conversation (579 bytes) Show as ASCII No delta times

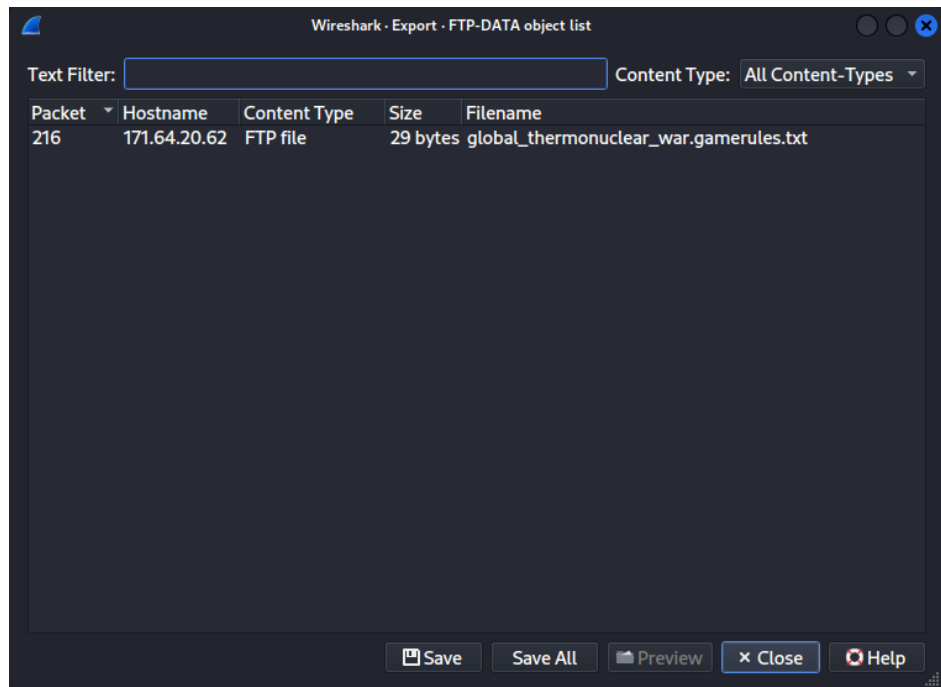
Find:

2. Find transferred file

Important line:

RETR global_thermonuclear_war.gamerules.txt

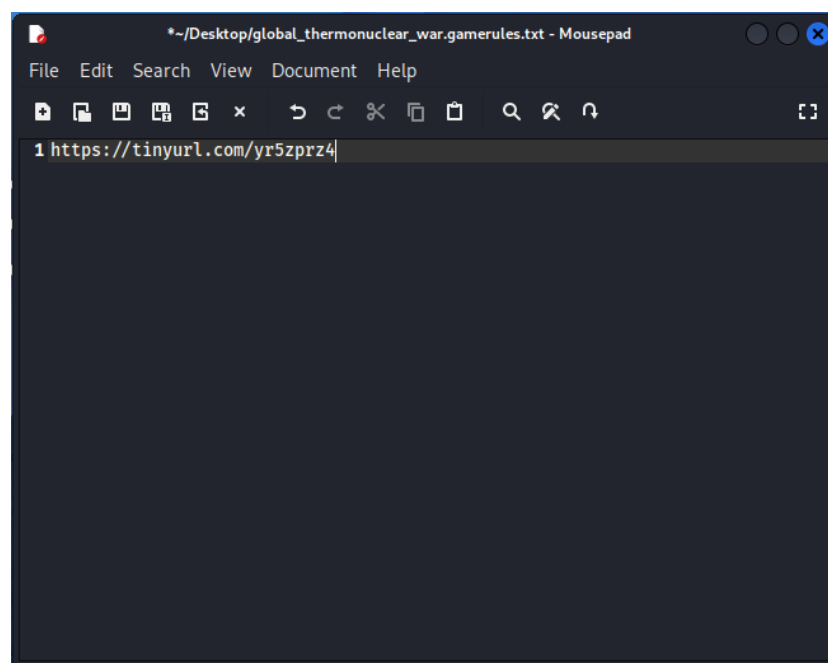
- A file was downloaded via FTP
- That file likely contains the flag
- Go to File - Export Object – Find FTP-DATA
- You can see global_thermonuclear_war.gamerules.txt and you save.



4. Open the file

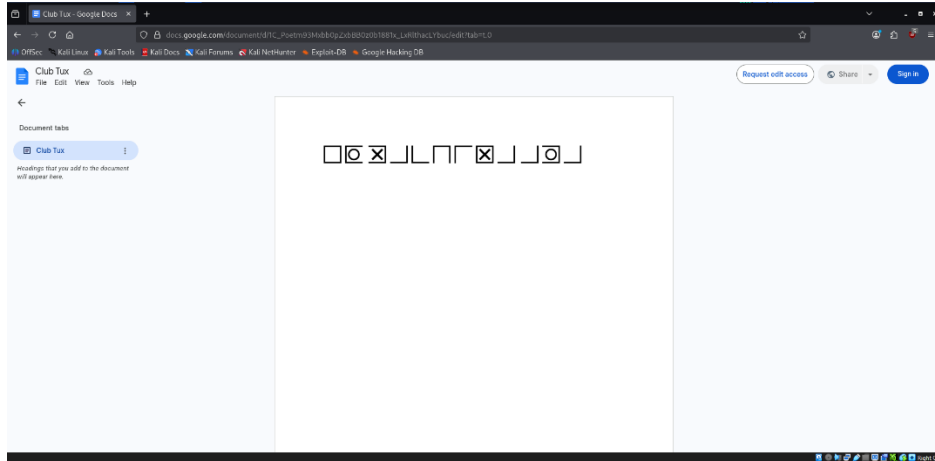
You got:

<https://tinyurl.com/yr5zprz4>



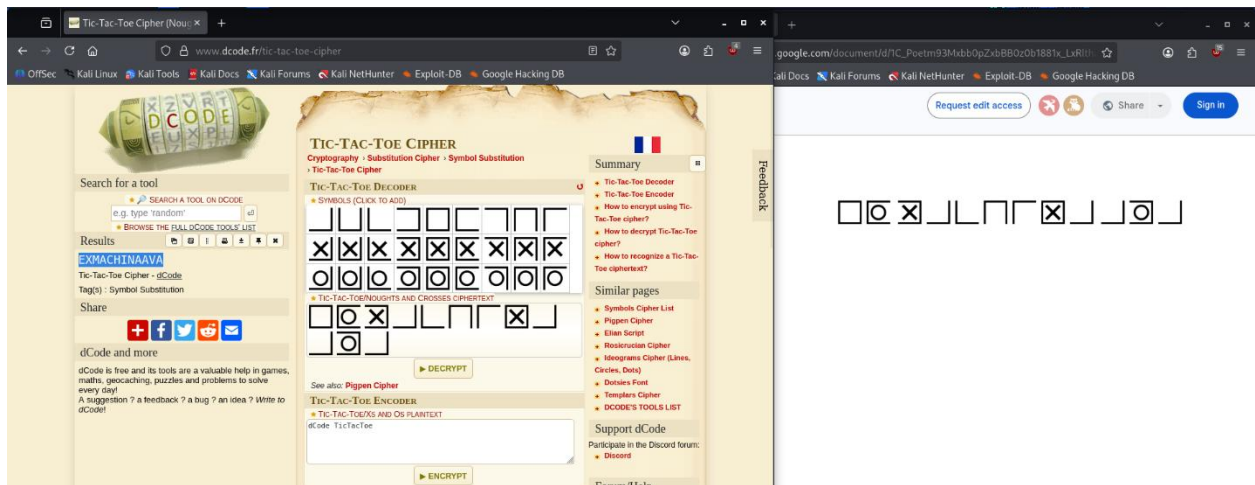
5. Open the TinyURL

- Open link URL
- You get some code or encoded text
- The code is Tic-Tac-Toe Cipher



6. Identify encoding type

- Open your browser and search <https://www.dcode.fr/tic-tac-toe-cipher>
- Tool used: dCode (Tic-Tac-Toe Cipher)
- You have to match the code and then you get the flag.
- The result gets **EXMACHINAABA**.



Question 3: Interpret an Nmap Output

PORT	STATE	SERVICE	VERSION
21/tcp	open	ftp	vsftpd 2.3.4
22/tcp	open	ssh	OpenSSH 5.3p1
80/tcp	open	http	Apache 2.2.8
139/tcp	open	netbios-ssn	
445/tcp	open	microsoft-ds	Windows 7 Professional 7601 Service Pack 1

Questions:

1. What can an attacker do with each port?

- Port 21 (FTP – vsftpd 2.3.4)
 - Upload and download files
 - Abuse anonymous login
 - Gain backdoor access (this version allows it)
- Port 22 (SSH – OpenSSH 5.3p1)
 - Crack password hashes via brute force
 - Execute commands remotely once access is gained
- Port 80 (HTTP – Apache 2.2.8)
 - Launch web-based attacks such as SQL injection, XSS, and directory traversal
 - Take control of the website
- Port 139 (NetBIOS)
 - View available shares and users
 - Gather information
- Port 445 (SMB – Windows 7 SP1)
 - Exploit SMB protocol flaws
 - Execute remote code injections (like EternalBlue)

2. What vulnerabilities are likely present based on the version?

- **vsftpd 2.3.4** → Backdoor vulnerability
- **OpenSSH 5.3p1** → Weak encryption / brute-force risk
- **Apache 2.2.8** → Outdated, many known exploits
- **SMB (Windows 7 SP1)** → EternalBlue (MS17-010)

3. Which one is the highest risk and why?

Port 445 (SMB)

- Allows **remote code execution without authentication**
- Widely exploited (e.g. WannaCry ransomware)
- Critical impact → full system takeover

4. What attack path can be built from this?

- Scan target → identify open ports
- Exploit **SMB (445)** → gain system access
- Use **NetBIOS (139)** → gather more info
- Use **FTP (21)** → upload malicious files
- Use **SSH (22)** → maintain persistent access
- Attack web server (80) → expand control

5. What should be the remediation?

- Update all outdated services
- Disable unused ports
- Patch SMB vulnerabilities (MS17-010)
- Use strong passwords & SSH key authentication
- Firewall restrictions
- Network segmentation

Question 4: Identify the OS (OS Fingerprinting) – TTL

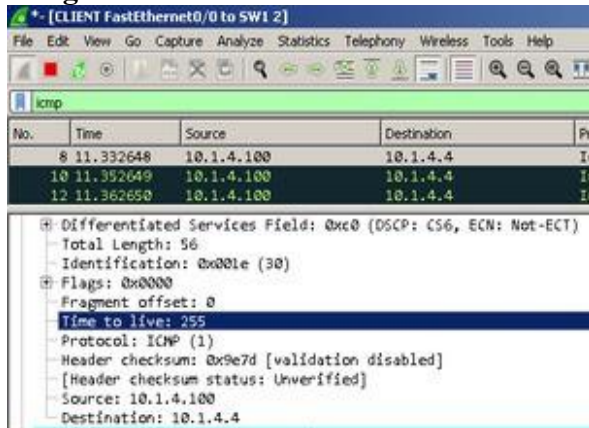
image 1

```
[sk@ostechnix ~]$ ping -c5 localhost
PING localhost(localhost (::1)) 56 data bytes
64 bytes from localhost (::1): icmp_seq=1 ttl=64 time=0.063 ms
64 bytes from localhost (::1): icmp_seq=2 ttl=64 time=0.057 ms
64 bytes from localhost (::1): icmp_seq=3 ttl=64 time=0.070 ms
64 bytes from localhost (::1): icmp_seq=4 ttl=64 time=0.067 ms
64 bytes from localhost (::1): icmp_seq=5 ttl=64 time=0.056 ms

--- localhost ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4096ms
rtt min/avg/max/mdev = 0.056/0.062/0.070/0.005 ms
```

Answer: OS Linux / Unix

Image 2



Answer: Network device (Router / Cisco)

Image 3:

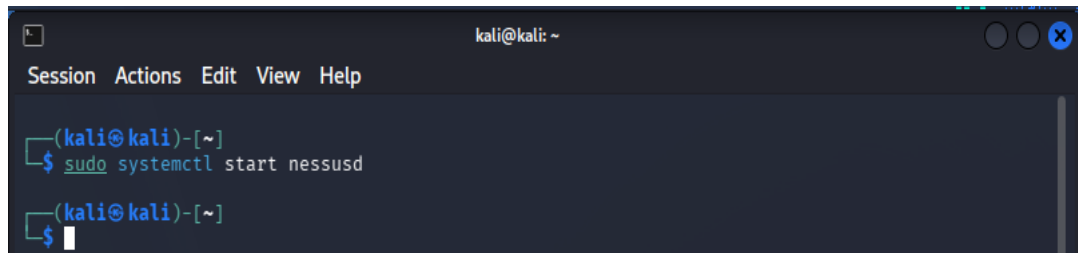
```
64 bytes from 192.168.122.239: icmp_seq=1 ttl=128 time=6.85 ms
64 bytes from 192.168.122.239: icmp_seq=2 ttl=128 time=0.510 ms
64 bytes from 192.168.122.239: icmp_seq=3 ttl=128 time=0.601 ms
64 bytes from 192.168.122.239: icmp_seq=4 ttl=128 time=0.541 ms
64 bytes from 192.168.122.239: icmp_seq=5 ttl=128 time=0.558 ms
```

Answer: OS Windows

Question 5: Analyse the Nessus file

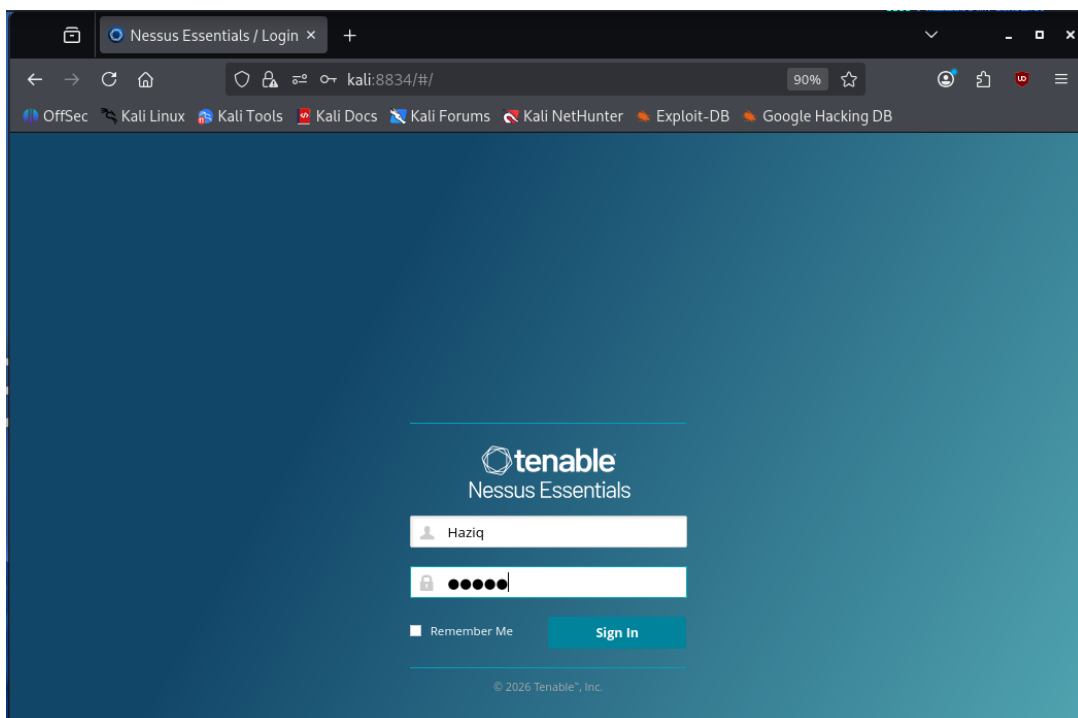
Upload to your nessus (Network_Scan.nessus) and analyse the files. Focus on critical or high findings that was identifies in analysis named “Ghostcat”.

- First you must set up you Nessus in kali
- After set up you must start or open your command in terminal
Sudo systemctl start nessusd

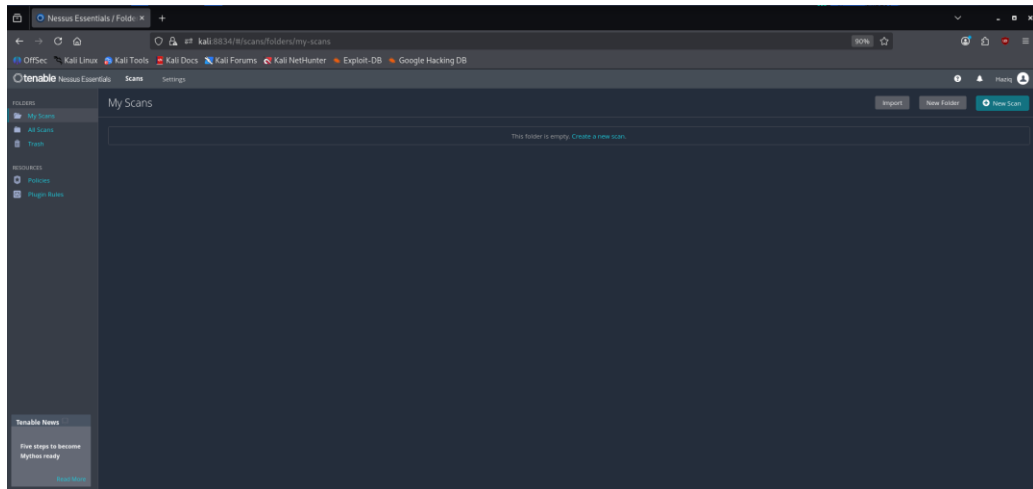


```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
$ sudo systemctl start nessusd  
(kali@kali)-[~]  
$
```

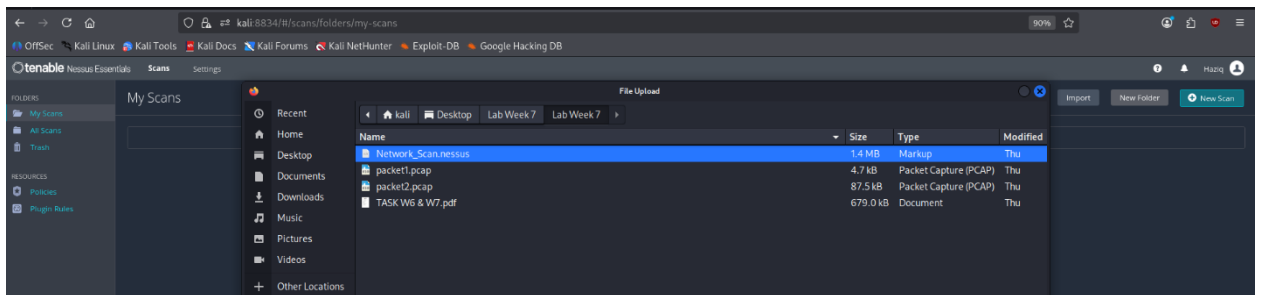
- Open your browser and type this command to open Nessud
<https://kali:8834> / <https://localhost:8834>
- Enter you name and password



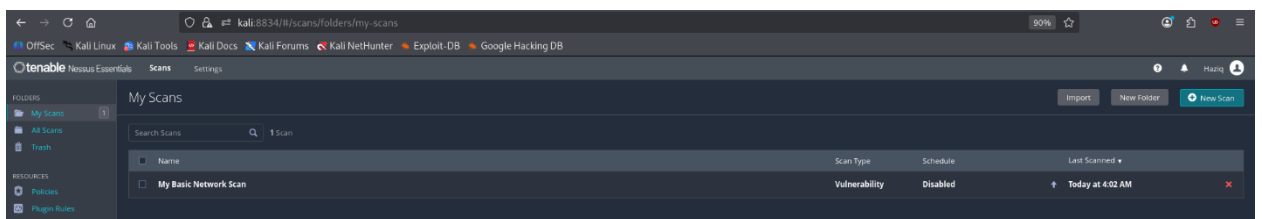
- This is a page for Nessus



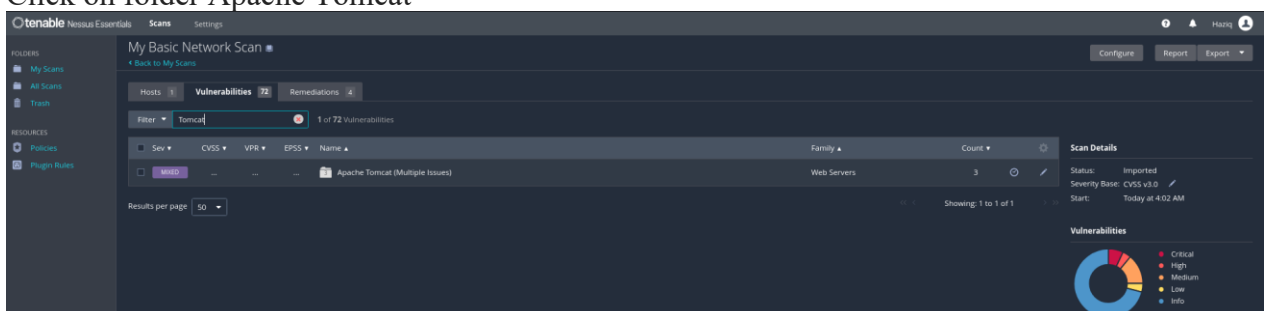
- First import file from Week 7
- Choose file Network_Scan.nessus
- And click Open



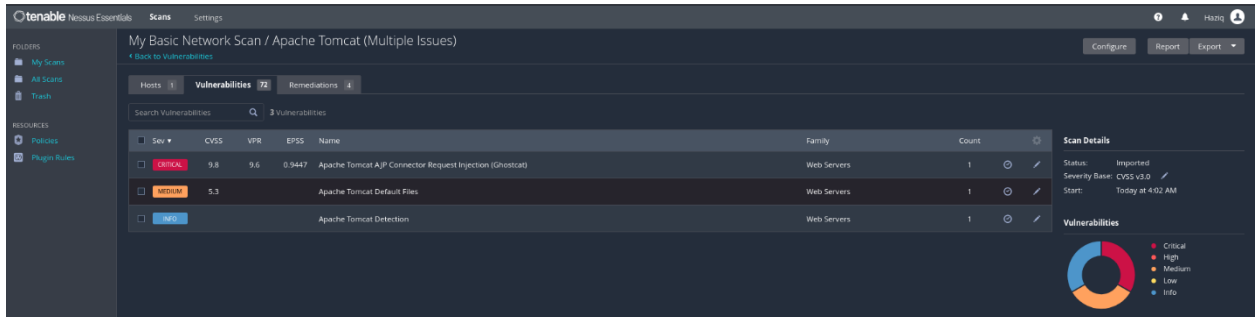
- This is you Scan



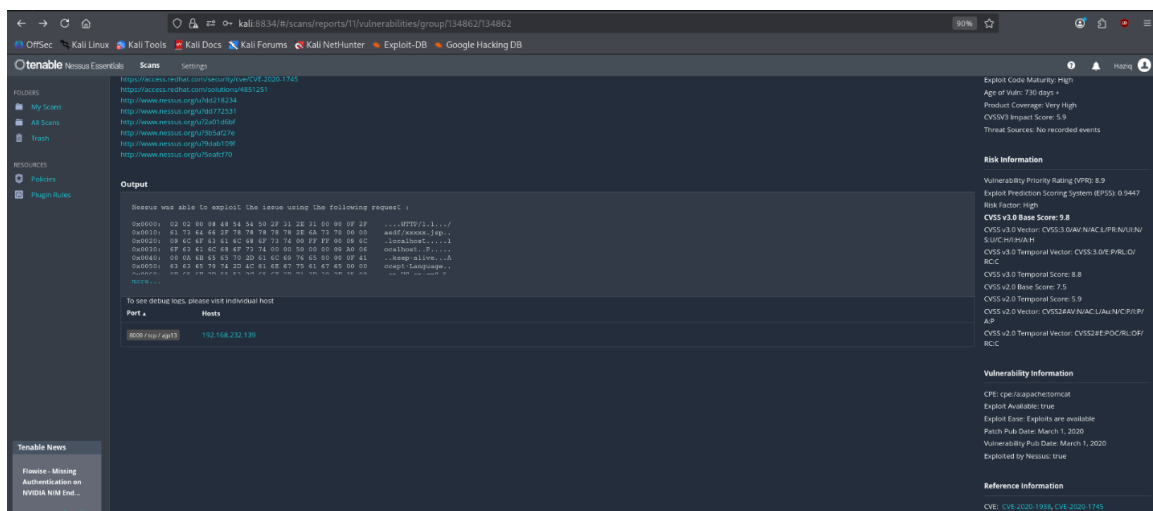
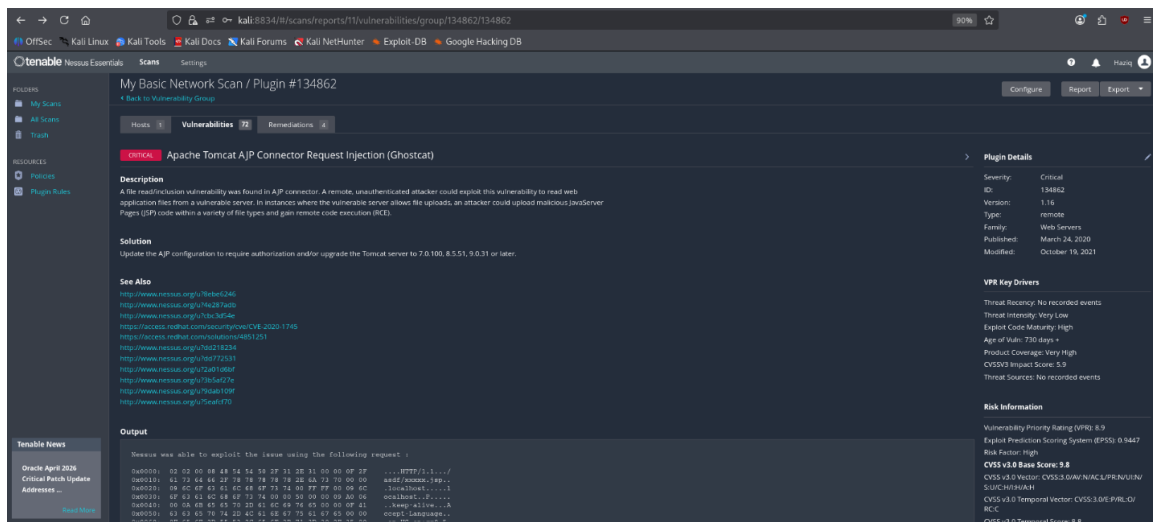
- After that you click on My Basic Network Scan
- Click on Vulnerabilities
- Search name Tomcat
- Click on folder Apache Tomcat



- You can see get 3 Vulnerabilities
 - CRITICAL
 - MEDIUM
 - INFO



- Click on CRITICAL Vulnerabilities
- Get name (Ghostcat)
- CVSS v3.0 Base Score 9.8
- Port 8009 / tcp /ajp12
- Hosts 192.168.232.139
- CVE : CVE-2020-1938, CVE-2020-1745



- 1. What is the affected Port number**
 - 8009
- 2. What is the Affected protocol**
 - AJP (Apache JServ Protocol)
- 3. What is the CVSS Score of vulnerability found**
 - 9.8 Critical
- 4. Can you find any exploit related to this vulnerability?**
 - File inclusion / file disclosure
 - /WEB-INF/web.xml
- 5. Find CVE for this vulnerability.**
 - CVE-2020-1938
 - CVE-2020-1745